

Assignment 3: Torrance Anonuevo 1008081549, Tanya Rustogi 1007828665

Part 2: Functional Dependencies, Decompositions, and Normal Forms

Question 1

(a) BCNF Violations

To find violations, since all attributes appear on the RHS, we compute the closures for each FD to see if the left-hand side (LHS) is a superkey.

- $L \rightarrow NQ$:
 $L^+ = \{L, N, Q\}$ (from $L \rightarrow NQ$)
 $L^+ = \{L, N, Q, S\}$ (from $NQ \rightarrow LS$)
 $L^+ = \{L, N, Q, S, O, P, R\}$ (from $S \rightarrow OPR$)
 $L^+ = \{L, M, N, O, P, Q, R, S\}$ (from $O \rightarrow M$)
Since L^+ contains all attributes, L is a superkey. This FD does **not** violate BCNF.
- $NQ \rightarrow LS$:
 $NQ^+ = \{L, N, Q, S\}$ (from $NQ \rightarrow LS$)
 $NQ^+ = \{L, N, Q, S, O, P, R\}$ (from $S \rightarrow OPR$)
 $NQ^+ = \{L, M, N, O, P, Q, R, S\}$ (from $O \rightarrow M$)
Since NQ^+ contains all attributes, NQ is a superkey. This FD does **not** violate BCNF.
- $S \rightarrow OPR$:
 $S^+ = \{O, P, R, S\}$ (from $S \rightarrow OPR$)
 $S^+ = \{M, O, P, R, S\}$ (from $O \rightarrow M$)
 S^+ does not contain $\{L, N, Q\}$. S is not a superkey. **Violation.**
- $O \rightarrow M$:
 $O^+ = \{M, O\}$ (from $O \rightarrow M$)
 O^+ does not contain all attributes. O is not a superkey. **Violation.**
- $MNR \rightarrow O$:
 $MNR^+ = \{M, N, R, O\}$ (from $MNR \rightarrow O$)
 MNR^+ does not contain all attributes. MNR is not a superkey. **Violation.**

(b) BCNF Decomposition

Step 1: Decompose using $O \rightarrow M$

Let $R_a = O^+ \cap R_1 = \{M, O\}$

Let $R_b = (R_1 - \{M\}) \cup \{O\} = \{L, N, O, P, Q, R, S\}$

Step 2: Check $R_a(MO)$

$O \rightarrow M$ is the only FD. O is a key for R_a . R_a is in BCNF.

Step 3: Check $R_b(LNOPQRS)$

Project S_1 onto R_b : $\{L \rightarrow NQ, NQ \rightarrow LS, S \rightarrow OPR\}$ ($MNR \rightarrow O$ and $O \rightarrow M$ are dropped since

$M \notin R_b$).

Keys of R_b : $\{L\}^+ = \{L, N, Q, S, O, P, R\} = R_b$ and $\{N, Q\}^+ = R_b$, so both $\{L\}$ and $\{N, Q\}$ are keys.

$S^+ \cap R_b = \{O, P, R, S\}$. S is not a superkey for R_b . **Violation.**

Decompose R_b using $S \rightarrow OPR$:

Let $R_c = S^+ \cap R_b = \{O, P, R, S\}$

Let $R_d = (R_b - \{O, P, R\}) \cup \{S\} = \{L, N, Q, S\}$

Step 4: Check $R_c(OPRS)$

$S \rightarrow OPR$ is the only projected FD. S is a key for R_c . R_c is in BCNF.

Step 5: Check $R_d(LNQ S)$

Projected FDs: $L \rightarrow NQ$ and $NQ \rightarrow LS$.

$\{L\}^+ = \{L, N, Q, S\} = R_d$ and $\{N, Q\}^+ = \{L, N, Q, S\} = R_d$. Both LHS are superkeys. R_d is in BCNF.

Final Relations:

1. $R_A(L, N, Q, S)$ with projected FDs: $\{L \rightarrow NQ, NQ \rightarrow LS\}$
2. $R_B(M, O)$ with projected FD: $\{O \rightarrow M\}$
3. $R_C(O, P, R, S)$ with projected FD: $\{S \rightarrow OPR\}$

(c) Dependency Preservation

This decomposition **does not** preserve all dependencies. The dependency $MNR \rightarrow O$ is lost because the attributes $\{M, N, R, O\}$ are not contained within a single relation in the final decomposition (M is in R_B , while N is in R_A , and R is in R_C).

(d) Chase Test

Row	L	M	N	O	P	Q	R	S
R_A	a_1	$b_{1,2}$	a_3	$b_{1,4}$	$b_{1,5}$	a_6	$b_{1,7}$	a_8
R_B	$b_{2,1}$	a_2	$b_{2,3}$	a_4	$b_{2,5}$	$b_{2,6}$	$b_{2,7}$	$b_{2,8}$
R_C	$b_{3,1}$	$b_{3,2}$	$b_{3,3}$	a_4	a_5	$b_{3,6}$	a_7	a_8

Applying $S \rightarrow OPR$: Rows R_A and R_C have the same S (a_8). We equate their O, P, R columns.

Row	L	M	N	O	P	Q	R	S
R_A	a_1	$b_{1,2}$	a_3	a_4	a_5	a_6	a_7	a_8
R_B	$b_{2,1}$	a_2	$b_{2,3}$	a_4	$b_{2,5}$	$b_{2,6}$	$b_{2,7}$	$b_{2,8}$
R_C	a_1	$b_{1,2}$	a_3	a_4	a_5	a_6	a_7	a_8

Applying $O \rightarrow M$: All rows now agree on O (a_4). We equate their M columns.

Row	L	M	N	O	P	Q	R	S
R_A	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8
R_B	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8
R_C	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8

The rows now consist of $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}$. Therefore, the decomposition is a lossless-join decomposition.

Question 2

(a) Minimal Basis

Step 1: Decompose.

- $AB \rightarrow C$
- $C \rightarrow A, C \rightarrow B, C \rightarrow D$
- $CFD \rightarrow E$
- $E \rightarrow B$
- $BF \rightarrow E, BF \rightarrow C$
- $B \rightarrow D, B \rightarrow A$

Step 2: Remove redundancy.

Check $CFD \rightarrow E$:

- Drop C ? $\{F, D\}^+ = \{F, D\}$. Cannot derive E . C is not redundant.
- Drop F ? $\{C, D\}^+ : C \rightarrow B; B \rightarrow A, D$. $\{C, D\}^+ = \{A, B, C, D\}$. Cannot derive E . F is not redundant.
- Drop D ? $\{C, F\}^+ : C \rightarrow B; BF \rightarrow E$. $\{C, F\}^+ = \{B, C, E, F\}$. D is **redundant**. Replace with $CF \rightarrow E$.

Check $BF \rightarrow E$ and $BF \rightarrow C$: $\{F\}^+ = \{F\}$ and $\{B\}^+ = \{A, B, D\}$, which contains neither E nor C . Neither B nor F can be dropped. Both remain.

Check $AB \rightarrow C$: $\{A\}^+ = \{A\}$ and $\{B\}^+ = \{A, B, D\}$, which doesn't contain C . Can't be dropped.

After Step 2: $\{AB \rightarrow C, C \rightarrow A, C \rightarrow B, C \rightarrow D, CF \rightarrow E, E \rightarrow B, BF \rightarrow E, BF \rightarrow C, B \rightarrow D, B \rightarrow A\}$

Step 3: Remove redundant FDs.

- $C \rightarrow A$: Without this FD, $\{C\}^+ : C \rightarrow B; B \rightarrow A$. A is still reachable. **Redundant**. Remove.
- $C \rightarrow D$: Without this FD, $\{C\}^+ : C \rightarrow B; B \rightarrow D$. D is still reachable. **Redundant**. Remove.
- $CF \rightarrow E$: Without this FD, $\{C, F\}^+ : C \rightarrow B; BF \rightarrow E$. E is still reachable. **Redundant**. Remove.
- $AB \rightarrow C$: Without this FD, $\{A, B\}^+ = \{A, B, D\}$. Cannot reach C . **Not redundant**. Keep.
- $C \rightarrow B$: Without this FD, $\{C\}^+ = \{C\}$. Cannot reach B . **Not redundant**. Keep.
- $E \rightarrow B$: Without this FD, $\{E\}^+ = \{E\}$. Cannot reach B . **Not redundant**. Keep.
- $BF \rightarrow E$: Without this FD, $\{B, F\}^+ : B \rightarrow A, D; BF \rightarrow C; C \rightarrow B$. $\{B, F\}^+ = \{A, B, C, D, F\}$. Cannot reach E . **Not redundant**. Keep.
- $BF \rightarrow C$: Without this FD, $\{B, F\}^+ : B \rightarrow A, D; BF \rightarrow E; E \rightarrow B$. $\{B, F\}^+ = \{A, B, D, E, F\}$. Cannot reach C . **Not redundant**. Keep.

- $B \rightarrow D$: Without this FD, $\{B\}^+ = \{A, B\}$. Cannot reach D . **Not redundant.** Keep.
- $B \rightarrow A$: Without this FD, $\{B\}^+ = \{B, D\}$. Cannot reach A . **Not redundant.** Keep.

Minimal Basis: $\{AB \rightarrow C, B \rightarrow A, B \rightarrow D, BF \rightarrow C, BF \rightarrow E, C \rightarrow B, E \rightarrow B\}$

(b) All Keys of R_2

Attributes F , G , and H never appear on any RHS in S_2 , so they cannot be derived and must appear in every key. We find the minimal set X such that $\{X, F, G, H\}^+ = R_2$. A and D can't be X since they are both reachable through B .

- $\{B, F, G, H\}$: $B \rightarrow A, D; BF \rightarrow C, E$; closure = $\{A, B, C, D, E, F, G, H\}$. ✓
Remove B ? $\{F, G, H\}^+ = \{F, G, H\}$. Cannot reach all attributes. **Minimal.** $\{B, F, G, H\}$ is a key.
- $\{C, F, G, H\}$: $C \rightarrow B; B \rightarrow A, D; BF \rightarrow C, E$; closure = R_2 . ✓
Remove C ? $\{F, G, H\}^+$ cannot reach B . **Minimal.** $\{C, F, G, H\}$ is a key.
- $\{E, F, G, H\}$: $E \rightarrow B; B \rightarrow A, D; BF \rightarrow C, E$; closure = R_2 . ✓
Remove E ? $\{F, G, H\}^+$ cannot reach B . **Minimal.** $\{E, F, G, H\}$ is a key.

No other minimal keys exist: any key must contain $\{F, G, H\}$, and the only starting points that reach all remaining attributes are B , C , or E .

All keys: $\{B, F, G, H\}$, $\{C, F, G, H\}$, $\{E, F, G, H\}$.

(c) 3NF Synthesis

Step 1: Create one relation per LHS group in the minimal basis.

LHS	RHS (combined)	Relation
AB	C	$S_1(A, B, C)$
B	A, D	$S_2(A, B, D)$
BF	C, E	$S_3(B, C, E, F)$
C	B	$S_4(B, C)$
E	B	$S_5(B, E)$

Step 2: Check whether any relation contains a key of R_2 .

The keys are $\{B, F, G, H\}$, $\{C, F, G, H\}$, $\{E, F, G, H\}$. None of the five relations contains G or H , so no relation contains a key. Add a relation for one key:

$$S_6(B, F, G, H)$$

Step 3: Remove subset relations.

- $S_4(B, C) \subseteq S_1(A, B, C)$? Yes, $\{B, C\} \subset \{A, B, C\}$. **Remove S_4 .**
- $S_5(B, E) \subseteq S_3(B, C, E, F)$? Yes, $\{B, E\} \subset \{B, C, E, F\}$. **Remove S_5 .**

Final 3NF Schema:

1. $S_1(A, B, C)$ with FDs: $AB \rightarrow C$ and $C \rightarrow B$; keys $\{A, B\}$ and $\{A, C\}$.
2. $S_2(A, B, D)$ with FDs: $B \rightarrow A$ and $B \rightarrow D$; key $\{B\}$.
3. $S_3(B, C, E, F)$ with FDs: $BF \rightarrow C$ and $BF \rightarrow E$; key $\{B, F\}$.
4. $S_6(B, F, G, H)$; key $\{B, F, G, H\}$.

(d) Does the Schema Allow Redundancy?

Yes, the schema can allow redundancy. 3NF synthesis guarantees dependency preservation and a lossless join, but it does not guarantee non-redundancy like BCNF does. Redundancy in 3NF is when a relation contains a FD $X \rightarrow Y$ where X is not a superkey but Y is a prime attribute.

In $S_1(A, B, C)$, the FD $C \rightarrow B$ holds. The keys of S_1 are $\{A, B\}$ and $\{A, C\}$, so B is prime. However, C alone is not a superkey of S_1 ($\{C\}^+ = \{B, C\} \neq \{A, B, C\}$). So S_1 is in 3NF but not BCNF. If two tuples share the same value of C , they must also agree on B , causing that B value to be repeated causing a redundancy.